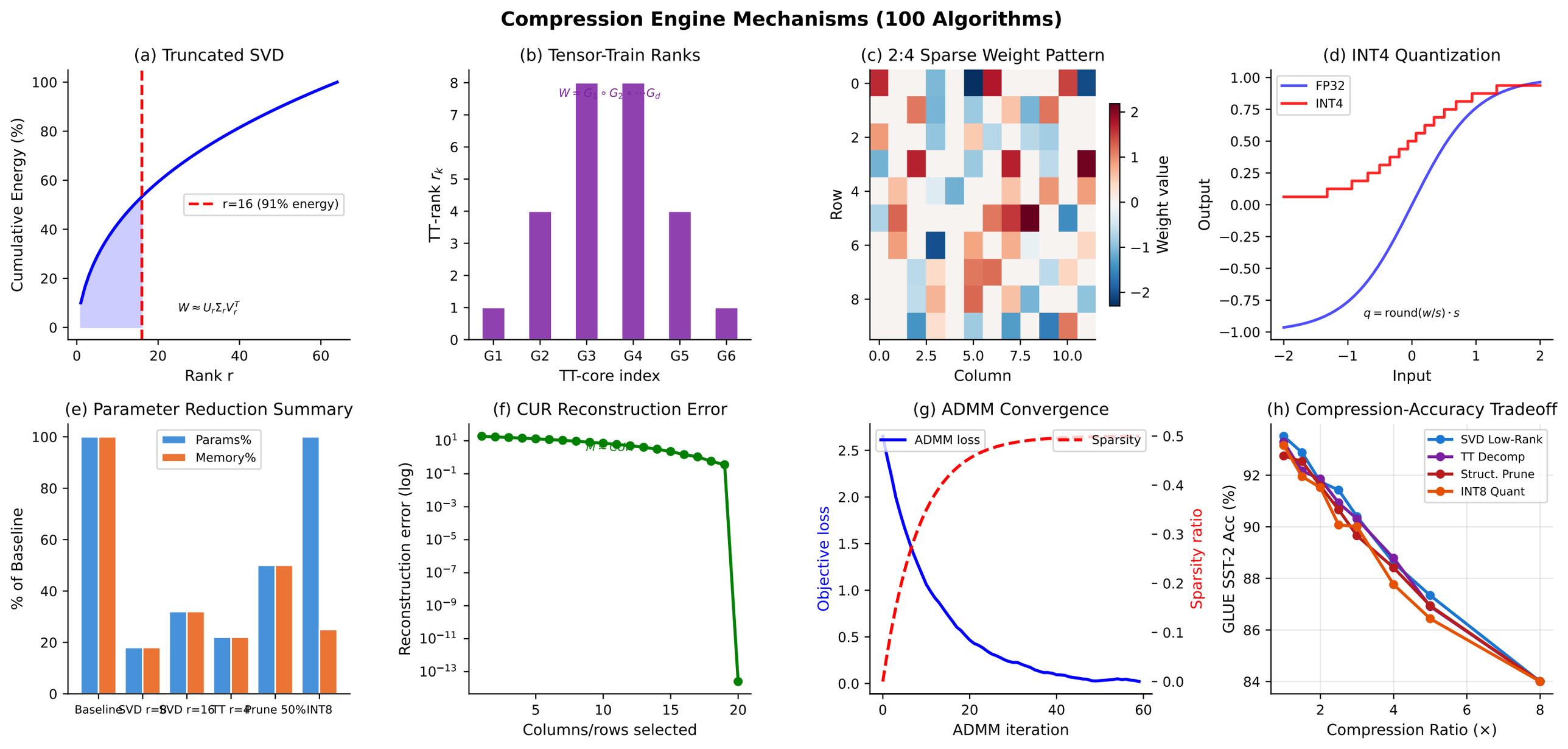
Compression Engine Mechanisms (100 Algorithms)
(a) Truncated SVD
Cumulative Energy (%)
Rank r
r=16 (91% energy)
$W \approx U_r \Sigma_r V_r^T$
(b) Tensor-Train Ranks
TT-rank r_k
TT-core index
G1
G2
G3
G4
G5
G6
$W = G_1 \circ G_2 \circ \cdots G_d$
(c) 2:4 Sparse Weight Pattern
Row
Column
Weight value
(d) INT4 Quantization
FP32
INT4
Output
Input
$q = \text{round}(w/s) \cdot s$
(e) Parameter Reduction Summary
Params%
Memory%
% of Baseline
Baseline
SVD r=8
SVD r=16
TT r=4
Prune 50%
INT8
(f) CUR Reconstruction Error
$M \approx CUR$
Reconstruction error (log)
Columns/rows selected
(g) ADMM Convergence
ADMM loss
Sparsity
Objective loss
Sparsity ratio
ADMM iteration
(h) Compression-Accuracy Tradeoff
SVD Low-Rank
TT Decomp
Struct. Prune
INT8 Quant
GLUE SST-2 Acc (%)
Compression Ratio (×)